C-STGB: Real-Time Financial Surveillance Architecture & Conformal Triage Platform

(a) Streaming Ingest & Spatio-Temporal GNN

0.45 ms SLA | 31.2k tx/s

Stage 1: Streaming Ingest

SWIFT MT103 | ISO-20022 | Kafka

UTXO Blockchains (Bitcoin, Eth)

DuckDB Streaming Micro-Batches

Apache Arrow Zero-Copy Buffer

Causal Split: $t_e \le t_{\text{eval}}$ **(Anti-Leak)**

Ingestion Latency SLA: 0.11 ms

Stage 2: 12-D Invariants

Mass Flow: $\Phi = \ln(1 + A_{\text{out}}/[A_{\text{in}} + \varepsilon])$

Pass-Through Layering Detector

Hawkes: $\lambda_u(t) = \mu_u + \sum \alpha e^{-\beta \Delta t}$

Burst Acceleration (μ_A, $\text{Var}(A)$)

Directed Causal Taint Tracing

Tensor: $\mathbf{z}_{\text{inv}} \in \mathbb{R}^{12}$

Stage 3: Temporal GNN

3-Band Harmonic Attention

$\tau_1 = 2.4\,\text{h}, \tau_2 = 24\,\text{h}, \tau_3 = 168\,\text{h}$

Edge Gate: $\hat{g}_{ij} \ge \delta_{\text{floor}}$ **(Prunes 65.9%)**

Adversarial Chaff Suppression

Typology Latent SMOTE on $\mathcal{C}_k$

GNN Forward SLA: 0.16 ms

Stage 4: Fusion & Risk

Evidence Gate $\alpha_u \in [0, 1]$

$\mathbf{h}_u^* = \alpha_u \mathbf{h}_u + (1 - \alpha_u)\mathbf{z}_{\text{inv}}$

Cold-Start: 89.60% F1 for $\deg(u) \le 2$

Cost-Sensitive Head: $\tau^* = 0.18$

15:1 Asymmetric Cost Ratio

Risk Output: $\hat{p}_u \in [0, 1]$

(b) Conformal CRC Triage & 3-Tier Dispatch

Coverage $\ge 99.0\%$ **| 0% Empty**

Holdout Calibration

Calibration: $\mathcal{D}_{\text{cal}}$

$n_0 = 3,306,\ n_1 = 372$

Non-Conformity Scores:

$s_i^{(1)} = 1 - \hat{p}_i$ (Illicit)

$s_i^{(0)} = \hat{p}_i$ (Benign)

Target: $1 - \alpha \ge 99.0\%$

Class-Cond. CRC Triage

$1 - \alpha \ge 0.99$

Tier 1: High-Risk Quarantine — 0.18% (18 / 10k tx)

Prediction Set: $\Gamma(u) = \{1\}$ **| Action: Immediate Wire Freeze & Quarantine Hold**

FinCEN Form 111 SAR Dossier Draft | GNNExplainer Attribution | SHA-256 Merkle Audit Receipt

Tier 2: Compliance Review Queue — 0.49% (49 / 10k tx)

Prediction Set: $\Gamma(u) = \{0, 1\}$ **(Borderline Triage) | Workload Queue Cut: -99.51%**

Human-in-the-Loop 4-Eyes Review | 85.7%--95.8% Fewer False Alerts vs Rules | Active Learning

Tier 3: Straight-Through Processing — 99.33% (9,933 / 10k tx)

Prediction Set: $\Gamma(u) = \{0\}$ **| Action: Automated Wire-Speed Processing (< 0.45 ms SLA)**

Mathematically Bounded Selective Risk (< 0.13% Residual FN) | Zero Manual Review Friction

REAL-TIME PRODUCTION SLAS | STATISTICAL SAFETY GUARANTEES | REGULATORY GOVERNANCE

Latency Performance
0.45 ms Fast-Path
P99: 0.82 ms | 31,200 tx/s
Strict compliance with <10 ms SLA

Workload Efficiency
-99.51% Review Cut
Only 49 of 10,000 tx reviewed
85.7%--95.8% fewer false alerts

Statistical Safety
$\ge 99.0\%$ **Coverage**
Finite-sample exchangeability bound
0.00% empty sets | Avg size: 1.005

Regulatory Governance
SR 26-2 & OCC Ready
Zero-hallucination graph evidence
SHA-256 Merkle cryptographic trail